ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

83000088 - Nuclear Energy In Surface Ships, Submarines, And Floating Artifacts

DEGREE PROGRAMME

08IN - Master Universitario En Ingenieria Naval Y Oceanica

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	83000088 - Nuclear Energy In Surface Ships, Submarines, And Floating Artifacts
No of credits	6 ECTS
Type	Optional/elective
Academic year ot the programme	Second year
Semester of tuition	Semester 3
Tuition period	September-January
Tuition languages	English
Degree programme	08IN - Master Universitario en Ingenieria Naval y Oceanica
Centre	08 - E.T.S. De Ingenieros Navales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Diana Cuervo Gomez (Subject coordinator)	Despacho	d.cuervo@upm.es	Sin horario. Published in ETSIN website

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CE4-(CE2) - Capacidad para analizar soluciones alternativas para la definición y optimización de las plantas de energía y propulsión de buques.Ability to analyze alternative solutions for the definition and optimization of ship power and propulsion plants.

CG3 - Que los estudiantes sepan comunicar sus conclusiones- y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades.That students know how to communicate their conclusions - and the ultimate knowledge and reasons that support them - to specialized and non-specialized audiences in a clear and unambiguous manner.

CG4 - (S1) Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CTUPM04 - (S5) Uso de la lengua inglesa. Los estudiantes establecen conversaciones con nativos sin tener problemas de comunicación adicionales tanto de forma oral como escrita.Use of the English language. Students establish conversations with native speakers without having additional communication problems both orally and in writing.

CTUPM06 - (S7) Comunicación oral y escrita. Los estudiantes transmiten conocimientos y expresan ideas y argumentos de manera clara, rigurosa y convincente, tanto de forma oral como escrita, utilizando los recursos gráficos y los medios necesarios adecuadamente y adaptándose a las características de la situación y de la audiencia.Oral and written communication. Students transmit knowledge and express ideas and arguments in a clear, rigorous and convincing manner, both orally and in writing, using the necessary graphic resources and media appropriately and adapting to the characteristics of the situation and the audience.

CTUPM13 - Trabajo en contextos internacionales. Los estudiantes son capaces de integrarse en un grupo o equipo, colaborando y cooperando con otros. Tienen la capacidad para trabajar con estudiantes de otras disciplinas y de aceptar la diversidad social y cultural.Work in international contexts. Students are able to integrate into a group or team, collaborating and cooperating with others. They have the ability to work with students from other disciplines and to accept social and cultural diversity.

3.2. Learning outcomes

RA56 - S1 Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo

RA69 - S5 - Uso de la lengua inglesa

RA70 - S8 - Respeto al medio ambiente

RA59 - S7 Comunicación oral y escrita.

RA57 - S2 Creatividad

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

This course aims to introduce students to the technology involved in nuclear installations in the marine environment, such as ship propulsion and offshore power generation.

It includes the fundamental concepts about Nuclear Physics and Reactor Physics that are needed to understand the behaviour of the nuclear reactors. Also, a detailed description of the Nuclear Technology is presented in order to understand the functioning of engineering equipment involved in this technology. In this way the students can understand how the nuclear reaction it is initiated, maintained and controlled by mean of different systems included in a nuclear reactor and also all the safety systems that are involved. Finally, specific characteristics of the nuclear installations for naval propulsion and off-shore energy production are also presented. Finally some fundamentals about nuclear submarines general arrangement, systems and buoyancy and stability will be studied.

4.2. Syllabus

1. Fundamentals
   1.1. Constitution of matter and nuclear energy
   1.2. Neutron interactions
   1.3. Nuclear fission
2. Reactor Physics
   2.1. The fission chain reaction
   2.2. Neutron moderation
   2.3. Neutron diffusion
   2.4. Reactivity coefficients
3. Nuclear reactors
   3.1. Fission nuclear reactors
   3.2. PWR nuclear reactors
   3.3. SMR reactors
   3.4. Fast reactors
4. Development of maritime nuclear propulsion
   4.1. Maritime nuclear propulsion history
   4.2. Military nuclear propulsion
   4.3. Civil nuclear propulsion
5. Characteristics of maritime nuclear propulsion installations
   5.1. Characteristics of maritime nuclear propulsion systems
   5.2. Nuclear propulsion: past, present and future reactors
   5.3. Safety and radiological protection
   5.4. Maritime safety for nuclear vessels
6. Other uses of nuclear energy off-shore
   6.1. Floating power plants
   6.2. Radioitotopes thermal generators
7. Fundamentals of nuclear submarines

7.1. General arrangement and services

7.2. Buoyancy, stability and drag resistance

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**Part I** Duration: 04:00 Lecture			
2	**Part I** Duration: 04:00 Lecture			
3	**Part I** Duration: 03:00 Lecture **Part II** Duration: 01:00 Additional activities			
4	**Part II** Duration: 04:00 Additional activities			
5	**Part II** Duration: 04:00 Lecture			
6	**Part II** Duration: 02:00 Lecture			**Midterm exam** Written test Progressive assessment Presential Duration: 02:00
7	**Part III** Duration: 04:00 Lecture			
8	**Part III** Duration: 03:00 Lecture **Part IV** Duration: 01:00 Lecture			
9	**Part IV** Duration: 02:00 Lecture **Part V** Duration: 02:00 Lecture			

10	**Part V** Duration: 04:00 Lecture			
11	**Part V** Duration: 03:00 Lecture **Part VI** Duration: 01:00 Research-based learning			
12	**Part VI** Duration: 02:00 Research-based learning			**Midterm exam** Written test Progressive assessment Presential Duration: 02:00
13	**Part VI** Duration: 04:00 Research-based learning			
14	**Part VII** Duration: 04:00 Research-based learning			
15				**Presentations** Group presentation Progressive assessment and Global Examination Presential Duration: 04:00
16				
17				**Exam** Written test Global examination Presential Duration: 04:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
6	Midterm exam	Written test	Face-to-face	02:00	40%	4 / 10	CE4-(CE2) CTUPM04
12	Midterm exam	Written test	Face-to-face	02:00	40%	4 / 10	CE4-(CE2) CTUPM04
15	Presentations	Group presentation	Face-to-face	04:00	20%	5 / 10	CG4 CTUPM06 CTUPM04 CG3 CTUPM13

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
15	Presentations	Group presentation	Face-to-face	04:00	20%	5 / 10	CG4 CTUPM06 CTUPM04 CG3 CTUPM13
17	Exam	Written test	Face-to-face	04:00	80%	5 / 10	CE4-(CE2) CTUPM13

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Exam	Written test	Face-to-face	04:00	80%	5 / 10	CE4-(CE2) CTUPM13
Presentations	Group presentation	Face-to-face	04:00	20%	5 / 10	CG4 CG3 CTUPM13 CTUPM06 CTUPM04

6.2. Assessment criteria

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
C.Ahnert, "Buques de Propulsión Nuclear", Ediciones ETSIN (2014)	Bibliography	Libro editado en la Escuela Técnica Superior de Ingenieros Navales con los apuntes del profesor
S. Glasstone, A. Sesonske, "Nuclear Reactor Engineering", Ed. Springer 1994	Bibliography	
Class materials	Web resource	Moodle
Material at Internet	Web resource	
Webcasts	Others	

8. Other information

8.1. Other information about the subject

This course is related with SDG7 "Affordable and clean energy" y con el SDG13 "Climate action"

The platforms that will be used to support teaching will be Moodle and Teams.